Uncertainty-aware CALPHAD–ML screening of heat-treatment intervals

Six-solute Ni-base superalloys · open mc_ni database · validated against six commercial alloys

WORKFLOW

1 Six-solute space, two linked CALPHAD calculations

matrix melting — liquid + fcc
solid state — γ′, MC, $M_{23}C_6$, σ, bcc
480-alloy LHS · 2 K solvus refinement

2 Uncertainty-aware surrogates

GP · PCE · NN · PCA-GP
calibrated prediction intervals
learning-curve-based budget

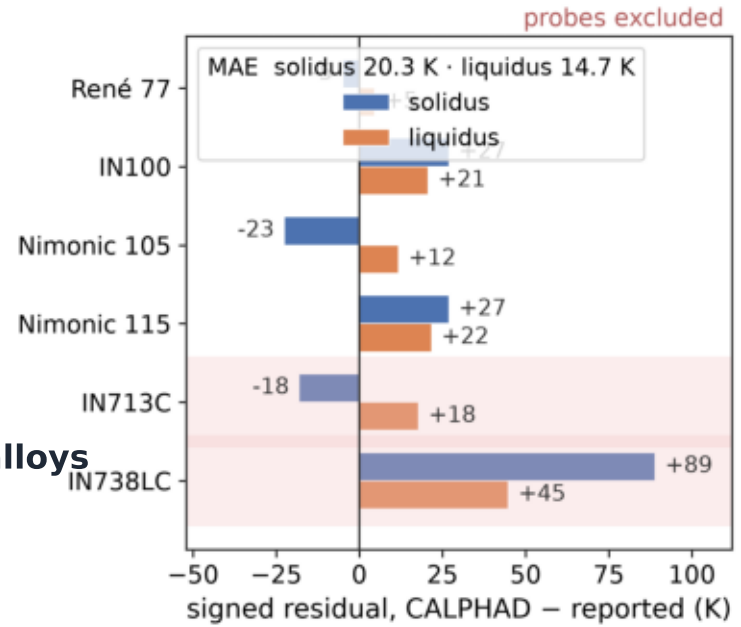
3 Validation against six commercial alloys

solidus 20.3 K · liquidus 14.7 K (MAE)
solvus cross-check −0.1 / +29.6 K
chemical — not geometric — domain

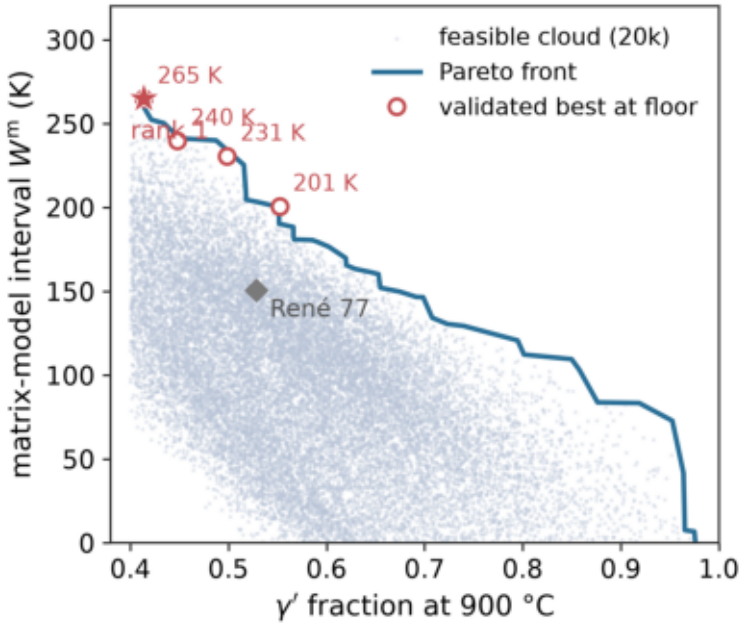
4 Inverse design with CALPHAD re-validation

120,000-candidate Sobol screen
Pareto front + γ′-floor sensitivity
σ-phase service screen at 750 / 900 °C

(a) Validation vs calorimetry



(b) Pareto design curve



✓ 264.8 K matrix-model interval — +114.3 K vs René 77, entirely by γ′ solvus depression

✓ P(gain > 0) = 98.1 % — bounded 87.9–100 % across priors, correlations, families

! Calculated ≠ measured window — a pre-registered paired DSC is the falsification test